

Introduction

Classically, a particle has position $x(t)$, momentum $p = mv$, and energy $E = \frac{p^2}{2m} + V(x)$, where m is mass, v is velocity and $V(x)$ is potential energy.

However, as shown by de Broglie's wave-particle duality, particles can be represented by waves in quantum mechanics, where the wavelength $\lambda = \frac{h}{p}$, with h being Planck's constant.

The wave associated with each particle can be represented in a wavefunction $\psi(x, t)$.

The Wavefunction ψ

Max Born's interpretation of the wavefunction is as follows:

$$|\psi(x, t)|^2 dx$$

...is the probability of a particle being at a location between x and $x + dx$.

This also shows that ψ is the probability amplitude and $|\psi|^2$ is the probability density; so, ψ is not a probability itself!

Furthermore, the total probability must equal 1.

i.e.
$$\int |\psi|^2 dx = 1$$

Deriving Schrödinger's equation

Start with a general sinusoidal wave: $\psi = A \sin\left(\frac{2\pi(x-ct)}{\lambda}\right)$

Differentiate twice with respect to x : $\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2}{\lambda^2} \psi$

De Broglie's formula: $\lambda = \frac{h}{p} \Rightarrow \frac{1}{\lambda^2} = \frac{p^2}{h^2}$

Formula for energy conservation: $E = \frac{p^2}{2m} + V$

Substitute in de Broglie's formula: $p^2 = 2m(E - V)$

Substitute again: $\frac{\partial^2 \psi}{\partial x^2} = -\frac{2m(E-V)}{\hbar^2} \psi$ where $\hbar = \frac{h}{2\pi}$

Rearrange: $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$

...which is the time-independent Schrödinger equation.

The time-dependent version: $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = i\hbar \frac{\partial \psi}{\partial t}$

Particle in a box

Imagine a particle trapped in a box with dimensions $0 < x < a$.

Thus, the potential energy $V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & x \leq 0, x \geq a \end{cases}$.

Inside the box:

$$V = 0$$

So, from the Schrödinger equation: $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$

or... $\frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0$ where $k^2 = \frac{2mE}{\hbar^2}$

Now, we try to solve the differential equation, starting with the general solution:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

At $x = 0$, the particle cannot exist. So, $\psi(0) = 0$.

Substituting in gives: $\psi(0) = A\sin(0) + B\cos(0) \Rightarrow B = 0$

$$\therefore \psi = A\sin(kx)$$

At $x = a$, the particle also cannot exist. So, $\psi(a) = 0$.

Substituting in gives: $0 = A\sin(ka)$

Which means... $ka = n\pi \Rightarrow k = \frac{n\pi}{a} \Rightarrow n = 1, 2, 3 \dots$

where n is the principal quantum number.

Then: $\therefore k^2 = \frac{2mE}{\hbar^2} \quad \& \quad k = \frac{n\pi}{a}$

$$\therefore \left(\frac{n\pi}{a}\right)^2 = \frac{2mE_n}{\hbar^2} \Rightarrow E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = \frac{h^2 n^2}{8ma^2}$$

...which shows that energy is discrete.

Now, normalise the wavefunction: $A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$

Integrate and substitute: $\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2} \Rightarrow A = \sqrt{\frac{2}{a}}$

Resulting wavefunction: $\therefore \psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$

By looking at the probability density function $|\psi|^2 = \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right)$, we can see that at $x = 0$ and $x = a$, the probability density is zero. Also, at higher n (i.e. energy levels), there would be more oscillations.

Heisenberg's Uncertainty Principle

The general formula for the expected value of a quantity $f(x)$ is as follows:

$$\langle f(x) \rangle = \int f(x) |\psi|^2 dx$$

The general formula for the standard deviation of a quantity $f(x)$ is as follows:

$$\Delta f(x) = \sqrt{\langle f^2(x) \rangle - \langle f(x) \rangle^2}$$

Looking at the average position in the box: $\langle x \rangle = \frac{a}{2}$

and the squared average position: $\langle x^2 \rangle = a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right)$

We get:
$$\Delta x = a \sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}$$

Given the momentum operator: $\hat{p} = -i\hbar \frac{d}{dx}$

We can find that: $\langle p \rangle = 0$ and $\langle p^2 \rangle = \frac{n^2\pi^2\hbar^2}{a^2}$

Thus:
$$\Delta p = \frac{n\pi\hbar}{a}$$

Combine what we have:
$$\Delta x \Delta p = \hbar \sqrt{\frac{n^2\pi^2}{12} - \frac{1}{2}}$$

As the smallest $\Delta x \Delta p$ can be is when $n = 1$ ($n \geq 1$), and $\sqrt{\frac{\pi^2}{12} - \frac{1}{2}} > \frac{1}{2} \dots$

$$\therefore \Delta x \Delta p > \frac{\hbar}{2}$$

This shows that the standard deviation of a particle's position and momentum can never be zero (which would mean that we know a particle's position or momentum to an exact value) and that their product must be greater than 5.27×10^{-35} Js.